ABDULRAUF OLASUNKANMI ABDULRAUF

Embedded Systems • Artificial Intelligence & ML • Control Systems • IoT

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PROFESSIONAL SUMMARY

A fresh graduate in Mechatronics Engineering (CGPA: 4.70/5.00) at the Federal University of Technology, Minna, with a proven track record in embedded systems design, machine learning implementation, and intelligent control systems. Demonstrated expertise spanning hardware-software integration, IoT sensor networks, PID control, LSTM-based predictive modelling, and deep learning with YOLO. Intern experience at NNPC and NITDA/NCAIR. Co-author of an ongoing peer-reviewed journal paper on self-balancing robotics. Seeking graduate trainee or full-time roles in embedded systems, AI/ML, or control engineering to deliver measurable impact across automation, IoT, and smart technology domains.

EDUCATION

B.Eng. Mechatronics Engineering | Federal University of Technology, Minna, Nigeria *2021 – Aug 2026*

- CGPA: 4.70 / 5.00 — consistently ranked among the top students in the department
- VC Academic Merit List: awarded consecutively from 100 Level through 500 Level
- Final Year Project: Design of a Multi-Sensor Soil Monitoring and LoRa Communication System for Precision Farming Applications with LSTM Model for Prediction Analysis
- Relevant Coursework: Embedded Systems, Control Systems, Robotics & Automation, Digital Electronics, MEMS, Machine Learning, Signal Processing
- Peer Tutor: recognised as exceptionally strong in Control Systems and Mathematics; provided academic support to the full class cohort across multiple semesters

WASSCE | Flora College, Ilorin, Kwara State *2014 – 2020*

- Passed with grades A–C in all subjects, including English Language and Mathematics

First School Leaving Certificate | Grace Foundation Nursery and Primary School, Ilorin *2005 – 2014*

INTERNSHIP EXPERIENCE

Engineering Intern — Automation Department | NNPC (Nigerian National Petroleum Corporation), Abuja*Jun 2025 – Jul 2025*

- Attached to the Automation Department of Nigeria's foremost national energy company
- Maintained and updated technical documentation including maintenance logs, equipment records, procurement documents, and engineering reports for automation systems
- Gained exposure to industrial-scale automation infrastructure, operational safety protocols, and departmental engineering workflows

Technical Intern — NCAIR Division | NITDA (National Information Technology Development Agency), Abuja*Jul 2025 – Nov 2025*

- Selected as a technical intern within NITDA's National Centre for Artificial Intelligence and Robotics (NCAIR), Nigeria's premier AI and robotics research institution
- Contributed to sports analytics dashboard development: executed full data science workflow (ingestion, cleaning, EDA, feature engineering, modelling, visualisation) on football datasets using Pandas, NumPy, Matplotlib, Seaborn, and Plotly
- Contributed to weather forecasting project: responsible for dataset sourcing from reliable sources and developing the data preprocessing and model training pipeline
- Strengthened applied skills across Python programming, machine learning, data science, and embedded systems through structured industry exposure

- Organisational internship certificate obtained; pending official release by NITDA

ENGINEERING PROJECTS

Multi-Sensor Soil Monitoring & LoRa Communication System (Final Year Project) | FUT Minna, Nigeria *2026*

- Designed and built a precision agriculture IoT system integrating soil moisture, temperature, pH, and NPK sensors with LoRa wireless communication for long-range data transmission to a remote gateway
- Implemented LSTM (Long Short-Term Memory) deep learning model for predictive analysis of soil health, enabling intelligent forecasting of moisture levels, irrigation scheduling, and crop yield indicators
- System hardware fully constructed; currently in active field testing phase
- Demonstrates full-stack IoT capability: sensor hardware, embedded firmware, wireless communication protocol, cloud data relay, and AI prediction layer

Self-Balancing Delivery Robot — Journal Paper In Progress | FUT Minna, Nigeria *2024 – Present*

- Engineered a three-wheel self-balancing robot using Arduino Uno, MPU6050 6-axis IMU, and L298N dual motor driver; implemented PID control algorithm achieving stable dynamic balancing on flat and mildly uneven surfaces
- Wrote and optimised embedded C++ firmware; calibrated IMU sensor for accurate real-time orientation measurement and feedback loop integration
- Co-authoring peer-reviewed journal paper with colleague Ameh Chris and faculty supervisors; paper documents hardware design, control architecture, PID tuning methodology, and experimental validation
- Group leader: directed a team of 5+ students through complete project lifecycle — concept, design, build, testing, and academic publication

Sun-Tracking Solar Panel System | FUT Minna — Collaborative (Team of 7), Nigeria *2024*

- Collaborated in a 7-member team to engineer an automatic sun-tracking solar panel system using dual LDR sensors to detect sunlight direction and servo/motor control to align panel for maximum energy capture
- System demonstrated continuous 180° directional alignment with the sun across full daylight hours, improving charging consistency vs. fixed-mount equivalent
- Served as group leader, coordinating sub-system integration, testing schedule, and final delivery

Sports Analytics Dashboard | NITDA / NCAIR — Team Project, Abuja *2025*

- Designed and delivered a multi-chart football analytics dashboard applying the complete data science pipeline from raw dataset to interactive visualisations
- Tools: Pandas, NumPy, Matplotlib, Seaborn, Plotly; surfaced player performance metrics, team comparisons, and trend analysis

Weather Forecast System | NITDA / NCAIR — Team Project, Abuja *2025*

- Contributed to a machine learning-based weather forecasting application; responsible for dataset collection from authoritative meteorological sources and preprocessing/training pipeline development

SmartTutor — AI-Powered Educational Website | Team Project *2024*

- Core logic developer for an AI-powered tutoring web platform; designed the intelligence layer, built dataset gathering and preprocessing pipelines, and contributed to front-end integration
- Delivered a functional educational tool combining web development (HTML/CSS) with AI-driven tutoring functionality

YOLO Transfer Learning — Custom Object Detection | Personal Research *2024*

- Applied transfer learning on pre-trained YOLO models using PyTorch to train domain-specific object detectors on custom labelled datasets
- Evaluated model performance across different training configurations; demonstrated proficiency in computer vision pipeline construction

Smart Home Automation Prototype | FUT Minna, Nigeria *2023*

- Prototyped a microcontroller-based home automation system enabling appliance control via serial communication

- Designed interface logic in Python and C; conducted manual input/output device testing and debugging

Personal Portfolio Website | Personal *2024 – Present*

- Developing a personal engineering portfolio website using HTML and CSS to showcase technical projects, publications, and skills (in progress)

TECHNICAL SKILLS

Languages: Python (Advanced), C++ (Intermediate), C# (Intermediate), Solidity (Intermediate), JavaScript (Intermediate), HTML/CSS (Advanced)

ML & AI: PyTorch, TensorFlow, Scikit-learn, YOLO (v5/v8 transfer learning), OpenCV, LSTM, CNNs, Regression, Classification, Decision Trees, Random Forest, Neural Networks

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Plotly, data preprocessing, EDA, feature engineering, model evaluation

Embedded: Arduino, ESP32, Raspberry Pi, STM32, NodeMCU, MPU6050 IMU, L298N, LoRa (SX1278), LDR sensors, servo/stepper motors, PID control, microcontroller interfacing

Cloud: Amazon Web Services (AWS) — foundational exposure

Software: MATLAB, Proteus (circuit simulation), AutoCAD, MATLAB, Microsoft Office Suite, Adobe Photoshop

Web & Frameworks: HTML, CSS, Flask (Python web applications)

Blockchain: Solidity, smart contracts, Web3 fundamentals, Ethereum basics

PUBLICATIONS & RESEARCH

Abdulrauf O.A., Ameh C., et al. (In Progress). Self-Balancing Delivery Robot: PID Control Architecture, Design, and Experimental Validation. [Target: Peer-Reviewed Engineering Journal — submission anticipated 2026]

CERTIFICATIONS & TRAINING

- Solidity and Blockchain Development — Industry Certification
- Embedded Systems Design — Industry Certification
- NITDA/NCAIR Technical Internship — Organisational Certificate (pending release; offer letter and documentation on file)
- Machine Learning — Supervised & Unsupervised Models: Regression, Classification, Decision Trees, Ensemble Methods
- Deep Learning with PyTorch — CNNs, LSTMs, Transfer Learning, YOLO object detection
- Web Development — HTML, CSS, Flask
- Web3 & Blockchain Technologies — Decentralised systems, Solidity, Ethereum
- Data Science Pipeline — End-to-end: collection, preprocessing, EDA, modelling, visualisation

LEADERSHIP & CONTRIBUTIONS

- Group Leader — Self-Balancing Robot: led 5+ person team through complete development lifecycle; co-authoring journal paper
- Group Leader — Sun-Tracking Solar Panel: coordinated 7-member team to successful delivery of automated solar tracking system
- Peer Tutor — Control Systems & Mathematics: provided academic support to entire class cohort; consistently sought out by peers for complex problem-solving
- VC Academic Merit List — 5 consecutive years (100 to 500 Level); one of the most sustained academic performance records in the department

LANGUAGES

English: Professional working proficiency (spoken and written)
Yoruba: Native / Full professional proficiency

INTERESTS

DIY Electronics & Robotics • Sports Analytics & Football • Mathematics Challenges • AI Research • Gaming & Simulations